Attention-Only Architectures Are Sufficient for General Language Understanding

VerRes Demo Team (test draft)

TEST DRAFT constructed for the VerRes HackPSU demo. Not a real manuscript; problems are planted on purpose.

Abstract

We argue that attention-only architectures solve general language understanding and review recent evidence.

1. Introduction

The Transformer replaced recurrence and convolutions with attention and achieved strong results on machine translation [1].

Vaswani et al. proved that attention-only models outperform recurrent models on every natural language processing task [1].

BERT demonstrated that pre-trained Transformers achieve human-level language understanding [2].

Residual connections made very deep networks trainable for image recognition [3].

2. Related Work

Recursive attention pruning has been shown to enable zero-shot protein folding with transformers [4].

Transformers have also been applied to tabular time-series forecasting with large gains [5].

3. Conclusion

Attention is therefore all that is needed for general language understanding.

References

[1] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems 30. arXiv:1706.03762

[2] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT 2019. arXiv:1810.04805

[3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. Proceedings of CVPR 2016. arXiv:1512.03385

[4] Kowalczyk, M., & Arendt, P. (2021). Recursive attention pruning for zero-shot protein folding. Journal of Computational Neuroscience Methods, 14(3), 211-229.

[5] Smith, J., & Okafor, T. (2019). Transformers for tabular time-series forecasting. Nature, 521(7553), 436-444. doi:10.1038/nature14539